

Method	$\mathbb{E}[Y Z = 1]$	$\mathbb{E}[Y Z = 0]$	Δ_Y	$\mathbb{E}_{\pi_a}[Z]$	$\mathbb{E}_{\pi_b}[Z]$	NIE
Causal Learner	0.410(30)	0.000	0.410(30)	0.550(20)	0.790(20)	0.098(10)
Cross-entropy	0.560(80)	0.130(30)	0.430(100)	0.270(40)	0.270(30)	0.011(8)

Table 1: Performance of causal agent on the DOORKEY environment and standard deviation of the mean.

Method	$\mathbb{E}[Y \Pi = a]$	$\mathbb{E}[Y \Pi = b]$
Causal Learner	0.240(20)	0.300(30)
Cross-entropy	0.230(20)	0.240(10)

Table 2: Reward obtained in the DOORKEY environment.

we observe a small increase in accumulated reward for the joint policy.

Conclusion

Since all causal conclusion depends on assumptions that are not testable in observational studies (?), it is natural to ask why we are justified in believing that a particular method finds a *causal* representation of the environment.

In work involving pre-defined variables, such justification can be found either through external distributional assumptions about the relationship between the structure of the model and the data it generates (?), or because the model belongs to a class of models of which so many examples have been observed, that meta-learning allows the structure to be identified (?).

In contrast, our work assumes a specific causal diagram which, along with the definition of Z and Y , ensures that Z is imbued with a natural interpretation as the mediating causal factor of the causal pathway from X to Y .

A more fundamental question is *why* a parsimonious model of causal knowledge, such as the SMOKING/CANCER example, is preferable to a detailed causal model of patient history. Indeed, if we adopt the view that a model should best fit the MDP (i.e. a generative view), it is difficult to see why parsimony would be preferred.

Although we do not claim to have a definite answer, our approach differentiates situations in which it can find causal knowledge from those in which it cannot, *without* referencing a generative/best fit criteria. Specifically, for a variable Z to be identified, it must be so relative to a policy π_a , as *something the agent could potentially do*, and is associated with a high reward ($\mathbb{E}[Y|Z = 1] > \mathbb{E}[Y|Z = 0]$), but it *might not do it under its baseline behavior* π_a . As a consequence, the policy π_a must be sub-optimal in order for the agent to determine a causal model.

At a glance, this may seem like a flaw in the method, but the idea that causation is ill-defined when one has a precise description of the physical world is old (?), and closely matches our common sense: When a child learns to ride a bicycle, a potential causal explanation for a fall, such as *steering too far from the center of the lane* ($Z = 0$) and hitting the curb ($Y = 0$), is only relevant in case the child *could have taken better* actions to keep near the center of the lane

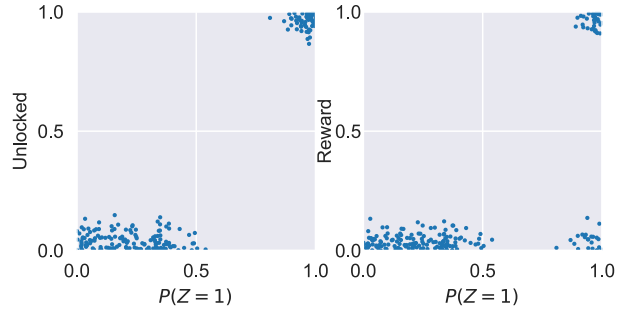


Figure 4: Scatter plot of $P(Z = 1)$ and (left) chance of unlocking the door, and (right) chance of successfully reaching the goal state. The causal variable Z appears to correspond to opening the door.

($Z = 1$). To put this in a different way, if the agent knows enough about the environment to have an optimal policy, a coarse-grained causal model cannot offer the agent any benefits because there are no policy decisions to improve.

This example hopefully clarifies a point made during review, namely how a choice of policy, $\Pi = \pi_a$ or $\Pi = \pi_b$ can act as a cause: The policy itself is not a cause, but rather the binary variable which denotes which policy is followed is treated as a cause.

We have argued that the NIE offers a novel way to define coarse-grained causal knowledge. In identifying a causal variable, our method learns a policy to manipulate it, and the variables are learned directly from experience *without* requiring specification of a generative process. We have shown the conditional expectations involved can be estimated using an n -step temporal difference methods, and that the method has convergence properties comparable to TD learning (but with worse constants). We found that the method was able to learn a causal variable which was both sensible and relevant for solving a task, and did so better than a natural (non-causal) alternative method. An independent experiment on a test-set indicated the causal representation is associated with an increased NIE, and that the causal representation is relevant for the task in that it resulted in a performance increase.

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